Name

xxx-xxx-xxxx • xxx@yahoo.com • linkedin.com/in/• https://github.com/

Skills and Technologies

- **Language**: C, C++, Python, Bash, Ladder Logic, Assembly(MIPS)
- **Embedded & Hardware:** ESP32, Raspberry Pi, Arduino, PLC integration, soldering & circuit assembly
- **Networking & Data Formats**: TCP/IP fundamentals, IP addressing & subnetting, Modbus, SPI/1-wire communication, JSON, XML
- **Tools & Platforms**: GitHub, Linux, Make, CMake, GCC, Visual Studio
- **Databases:** Firebase Firestore, MongoDB, SQL

Professional Experience

Embedded Software Engineering Intern **May 2025-Present**
Company

- Developed **Python**-based data acquisition system **integrating 50+ industrial PLCs** via **Modbus/TCP** protocol, emphasizing **reliable data transfer**, fault handling, and continuous operation across **distributed production environments**
- Programmed **ladder logic** for industrial **PLC** applications, implementing timer-based control sequences, I/O logic, and safety interlocks for a new commercial product line **generating $100,000+ in revenue**
- Developed **Java** applications used by **thousands** of clients using **object-oriented design principles**, emphasizing modularity, maintainability, and clear abstractions
- Collaborated in an **Agile** development environment, contributing across the **full software development lifecycle,** including development, build, testing, and deployment using **Git** and **CI workflows**
- Validated **PLC** outputs using a combination of **manual testing and automated test routines** to ensure correct I/O behavior and system reliability

Data Science Intern **March 2025 – May 2025**
Company

- Performed in-depth exploratory data analysis using **Python** and **R** to uncover statistical patterns, anomalies, and correlations in structured and unstructured datasets, directly informing **3+** new investment strategies
- Contributed to the development of a scalable **Python-based** back testing infrastructure, automating the evaluation of 50+ quantitative strategies across multiple asset classes

Projects

Dynamic Memory Device Driver | Source Code

- Engineered a **Linux kernel module** for a **character device driver** in **C,** implementing **dynamic memory management** with hierarchical data structures, achieving zero data corruption across **1000+ concurrent read/write** operations through **semaphore-based synchronization**
- Implemented standard **POSIX** file operations (open, read, write, llseek, release) at **kernel level** with proper resource cleanup and error handling, enabling seamless integration with **Linux VFS layer**
- Applied core **operating system concepts** including **kernel/user space** separation, synchronization, and **resource management**
- Created **shell scripts(Bash)** to **automate** building, loading, unloading of the kernel module and tested via **command line**

Weather Monitoring Terminal — ESP32 Microcontroller | Source Code | Deployed site

- Designed and deployed an end-to-end weather monitoring system **serving 2,000+** university students and faculty, **interfacing 4 sensors** (solar, rain, wind, temperature) with **ESP32 microcontroller** to provide real-time meteorological data for statistics courses and research programs
- Developed **embedded firmware in C/C++** for ESP32 platform implementing interrupt-driven sensor acquisition via **SPI and 1-Wire** communication protocols, achieving **24/7 system uptime** with real-time data processing and transmission
- Implemented secure **over-the-air (OTA)** firmware update functionality for **ESP32** devices, **eliminating the need for on-site reflashing** and enabling remote deployment of bug fixes and feature updates

EDUCATION

University — *B.S. in Computer Science & Accounting* ***May 2026***
GPA: 3.75
Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Design, Computer Architecture, Software Development, Operating Systems, AI Robotics, Information Security